

AISHWARYA KAGADAL

9110647692 | kagadalaishwarya@gmail.com | www.linkedin.com/in/aishwarya-kagadal

CAREER OBJECTIVE

Passionate Software Engineer skilled in C++ and Python with strong foundations in Data Structures, Algorithms, and Object-Oriented Programming, seeking to build scalable, cloud-based solutions and contribute to innovative projects at Salesforce.

WORK EXPERIENCE

Embedded and IOT Developer – Uilatech LLP, Belagavi Jan 2025 – Apr 2025

- Engineered BLE-based OTA firmware update system using Zephyr RTOS and MCUBoot, achieving 100% reliable firmware rollback and update success.
- Optimized C++ firmware for power and memory efficiency, enhancing system response by 20%.
- Collaborated cross-functionally to debug and integrate hardware-software modules, improving reliability.

Internship Trainee - Harsha Sugar Pvt Ltd, Saundatti Dec 2023 – Jan 2024

- Studied PLCs, sensors, and SCADA systems used in large-scale industrial automation.
- Analyzed sensor-actuator integration for real-time process monitoring and fault detection.

Summer Internship Trainee – Gogte Institute of Technology Oct 2022 – Nov 2022

- Programmed Arduino Uno for sensor data acquisition and control using C.
- Built IoT prototypes for temperature and motion detection, improving peripheral interfacing skills.

EDUCATION

Bachelor of Engineering (B.E.) – Electronics and Communication Engineering
K.L.S Gogte Institute of Technology, Belagavi 2021 - 2025
CGPA: 8.59

Pre-University Course (PUC) - PCMS
KLE's Prerana PU College, Hubli 2019 - 2021
Percentage: 99%

Secondary School (10th)
Alva's English Medium High School, Moodbidri, Dakshina Kannada 2018 - 2019
Percentage: 97.28%

SKILLS

- Programming** - C++, Python, SQL, Data Structures & Algorithms, Object Oriented Programming.
- Core Concepts** - Data Analytics, Statistics, Digital Electronics, Microcontrollers & Sensors, Embedded & IoT, Computer Networks, System Design.
- Frameworks & Tools** - CST Studio, MATLAB/Simulink, Cisco Packet Tracer, Visual Studio, MySQL, MS EXCEL.
- Operating System** – Zephyr RTOS (Basic), Linux(Basic), Windows.

PROJECTS

ATM software implementation using C

- Developed a functional ATM software system including user authentication, balance inquiry, cash withdrawal, and deposit functionality.
- Developed a **menu-driven banking simulation** including authentication, transactions, and data validation.
- Applied **modular programming** and **file I/O** concepts for persistent storage.

C++ Library Management System

- Developed a Library Management System in C++ using Object-Oriented Programming concepts to manage book records, user registration, and transaction tracking with efficient file handling.
- Implemented search, sort, and update functionalities using data structures like maps and vectors, improving data retrieval speed and system usability.

Design and analysis of microstrip array antenna for WIFI Applications

- Designed and simulated a microstrip array antenna using CST Studio to enhance gain and directivity for Wi-Fi applications, incorporating beamforming for improved signal control.

Firmware update and firmware testing of aeroponic automation system.

- Designed and implemented a secure OTA firmware update and rollback system using BLE and MCUboot for an aeroponic automation system on nRF52840.
- Conducted comprehensive firmware testing to ensure system stability and fault recovery during updates.

CERTIFICATIONS

NPTEL Certification – Joy of computing using python.

Scalar Certification - Learn Essentials C++ programming language.

Deloitte Data Analytics Job Simulation – Forage (Virtual Internship)

Cisco Networking Academy – Introduction to IoT and Digital Transformation

Cisco Networking Academy – Introduction to Packet Tracer

ACHIEVEMENTS

HackerRank 5 Star (Gold Badge) – SQL

Cisco Networking Academy Badge – Introduction to IoT

Cisco Networking Academy Badge – Introduction to Packet Tracer

Cisco Networking Academy Badge – Networking Basics

PERSONAL INTEREST

- **Poetry** (Kannada language)
- **Cooking**